OECD Education Policy Perspectives 125

The impact of digital technologies on students' learning in Ireland

Key considerations from peers' experience

OECD
BETTER POLICIES FOR BETTER LIVES

This work is published under the responsibility of the Secretary-General of the OECD.

The opinions expressed and arguments employed herein do not necessarily reflect the official views of the Member countries of the OECD.

This document, as well as any data and map included herein, are without prejudice to the status of or sovereignty over any territory, to the delimitation of international frontiers and boundaries and to the name of any territory, city or area.

The statistical data for Israel are supplied by and under the responsibility of the relevant Israeli authorities. The use of such data by the OECD is without prejudice to the status of the Golan Heights, East Jerusalem and Israeli settlements in the West Bank under the terms of international law.

Please cite this publication as:
OECD (2025), "The impact of digital technologies on students' learning in Ireland: Key considerations from peers' experience", *OECD Education Policy Perspectives*, No.125, OECD Publishing, Paris, https://doi.org/10.1787/38f23920-en.

Photo credits: Cover © Khakimullin Aleksandr / Shutterstock

Corrigenda to OECD publications may be found at: https://www.oecd.org/en/publications/support/corrigenda.html.

© OECD 2025

Attribution 4.0 International (CC BY 4.0)

This work is made available under the Creative Commons Attribution 4.0 International licence. By using this work, you accept to be bound by the terms of this licence (https://creativecommons.org/licenses/by/4.0/).

Attribution – you must cite the work.

Translations – you must cite the original work, identify changes to the original and add the following text: In the event of any discrepancy between the original work and the translation, only the text of original work should be considered valid.

Adaptations – you must cite the original work and add the following text: This is an adaptation of an original work by the OECD. The opinions expressed and arguments employed in this adaptation should not be reported as representing the official views of the OECD or of its Member countries.

Third-party material – the licence does not apply to third-party material in the work. If using such material, you are responsible for obtaining permission from the third party and for any claims of infringement.

You must not use the OECD logo, visual identity or cover image without express permission or suggest the OECD endorses your use of the work.

Any dispute arising under this licence shall be settled by arbitration in accordance with the Permanent Court of Arbitration (PCA) Arbitration Rules 2012. The seat of arbitration shall be Paris (France). The number of arbitrators shall be one.

Abstract

The integration of digital technologies is reshaping education systems, offering both opportunities and challenges. In Ireland, the Department of Education and Youth has recognised the need for action to ensure inclusiveness and responsiveness in the digital age. To support this, the OECD Secretariat and Ireland's Department of Education and Youth organised an international peer-learning event on 25 November 2024 in Dublin. This work is part of the project *Resourcing School Education: Policies for the Digital Transformation of Education and Future-Readiness of Teachers*.

The event gathered experts from Estonia, France and Sweden to share experiences and strategies with Ireland's Department of Education and Youth. Discussions focused on measuring the impact of digital tools, ensuring equitable access, and fostering digital competences among students and teachers. This document summarises the discussion and insights, structured in five sections: opportunities and challenges, measuring impact, access and competence, success factors and conclusions.

Acknowledgements

This document and the related peer-learning event were commissioned by the Department of Education and Youth in Ireland in the context of the OECD project *Resourcing School Education: Policies for the Digital Transformation of Education and Future-Readiness of Teachers*, which is included in the 2025-26 Programme of Work of OECD's Education Policy Committee.

The document was prepared by Jose-Luis Alvarez-Galvan and Beatrice Bottura (OECD Secretariat), based on contributions from international experts who participated in the peer-learning event: Margus Pedaste (Estonia), Peter Karlberg (Sweden), and Maxime Barilleau (France). The OECD Secretariat gratefully acknowledges the valuable support of Susan Mulhall, Clare Connolly, and Edel Martin from the Department of Education and Youth in Ireland in organising the event in Dublin and contributing to the production of this document. The overall guidance and feedback on drafts provided by Andreas Schleicher (Director of Education and Skills) and Paulo Santiago (Head of Division, Policy Advice and Implementation) are gratefully acknowledged. Christina Mitrakos and Sophie Limoges (OECD Secretariat) supported the preparation of this document for its publication. Last but not least, this document has also benefited from the valuable input from delegates to the OECD Group of National Experts on School Resources (GNE-SR), which oversees the project.

Table of contents

Abstract 3

Acknowledgements 4

Executive summary 6

1 Opportunities and challenges of digital technologies in education: OECD and Ireland's overview 7

2 Measuring the impact of digital technologies in education 10

3 Ensuring quality, competence and equitable access in digital education 18

4 Key factors for successful outcomes in digital education 23

5 Conclusions and main considerations 28

Annex A. Agenda of the international peer-learning event 30

Annex B. Digital education initiatives mentioned in this paper 31

References 34

Notes 37

FIGURES

Figure 1. Students score better in mathematics when they spend moderate time using devices for learning at school but not when engaging in leisure activities 7
Figure 2. Underlying principles and values of Estonia's Education 2035 Strategy 12
Figure 3. The six steps of the Digital Learning Planning Cycle 15
Figure 4. The Primary Curriculum Framework's core competencies 16
Figure 5. Key themes of the Literacy, Numeracy and Digital Literacy Strategy 2024-2033 26

TABLES

Table B.1. Digital education initiatives in Estonia 31
Table B.2. Digital education initiatives in France 31
Table B.3. Digital education initiatives in Ireland 32
Table B.4. Digital education initiatives in Sweden 33

BOXES

Box 1. Balancing benefits and distractions, the dual impact of digital technologies on student learning 9

Executive summary

Digital technologies are transforming education systems, offering new opportunities but also posing challenges. To support its digital strategy, Ireland's Department of Education and Youth partnered with the OECD Secretariat to host an international peer-learning event in Dublin (25 November 2024). Experts from Estonia, France, and Sweden shared experiences on the impact of digital tools on student learning.

Informed by the peer-learning event, OECD evidence and national policy experiences, this document presents key insights across five areas: 1) opportunities and challenges, 2) measuring impact, 3) ensuring quality and equity, 4) key success factors, and 5) conclusions and main considerations.

Opportunities and challenges. PISA 2022 shows that moderate use of digital tools for learning is linked to better academic outcomes and a stronger sense of belonging. Irish students perform above average in core subjects and are confident in using digital platforms. Yet, distractions remain a concern, and equity gaps persist. Participants agreed that successful digital integration requires clear pedagogical intent, support for teachers, and targeted policy measures.

Measuring impact. Measuring the effects of digital tools on learning is complex. While PISA and ICILS provide useful data, they do not offer conclusions regarding causality. Countries are thus turning to mixed methods, including surveys, classroom observation, longitudinal research, and RCTs. Estonia's DigiEfekt study highlights that the type of digital use matters more than frequency. France's EduPilote and Sweden's national efforts aim to connect technology use with academic outcomes. Ireland's Digital Learning Framework provides a foundation for progress, and there is potential to strengthen it with more robust monitoring tools and data-driven approaches

Ensuring quality, competence and equity. Digital education must address issues related to infrastructure, teacher capacity, and equity. Ireland's Digital Strategy, DEIS programme, and ICT Grant Scheme target these issues. Teacher training is supported through Oide, and digital skills are included in initial teacher education. France offers digital skills certification via Pix, Estonia promotes coding through ProgeTiger, and Sweden supports local innovation, despite regional disparities. Ireland's Webwise programme also fosters digital citizenship and safety.

Key factors for success. Participants highlighted key enablers:

- Equitable access to technology
- Purposeful pedagogical integration
- Strong teacher training
- Stakeholder engagement
- Monitoring and evaluation

Ireland's strategy aligns with these principles, but there are opportunities to enhance data use and evaluation of learning outcomes to further strengthen its approach.

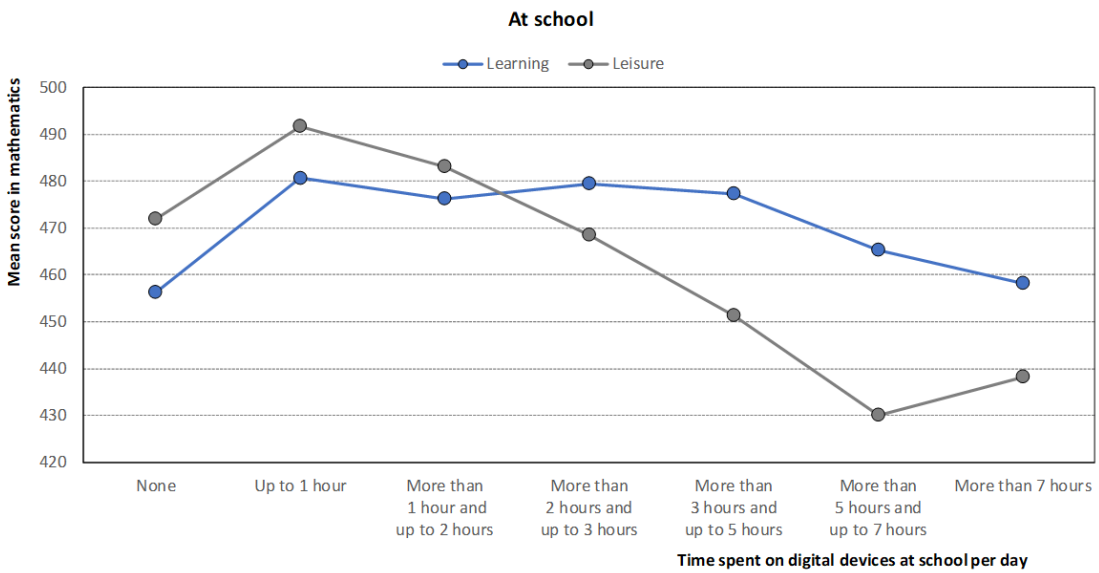
Final considerations. Ireland is making progress in digital education. To maximise impact, it should continue to improve monitoring systems, ensure equitable access and support teachers. Experiences from other countries might assist Ireland in this endeavour.

1 Opportunities and challenges of digital technologies in education: OECD and Ireland's overview

During the event, participants discussed the integration of digital technologies in teaching and learning worldwide, sparking considerable debate regarding their impact on student learning outcomes. As a basis for this discussion, the OECD team outlined some available evidence. Results from the 2022 edition of the Programme for International Student Assessment (PISA) highlight a complex relationship between digital device use and academic achievement (see Box 1). For example, PISA 2022 findings indicate that a moderate use of digital devices for educational purposes is associated with improved student performance and a stronger sense of belonging at school. Specifically, as shown in Figure 1, students who spend a moderate amount of time engaging with digital tools for learning activities tend to score better in mathematics compared to those who do not use these tools at all (OECD, 2024[1]). This evidence indicates the potential benefits of integrating digital technologies into educational settings, particularly when this is done judiciously, moderately, and with clear pedagogical intent[1].

Figure 1. Students score better in mathematics when they spend moderate time using devices for learning at school but not when engaging in leisure activities

Based on students' reports (OECD average).



Note: Differences between categories are all statistically significant.
Source: OECD, PISA 2022 Database, https://webfs.oecd.org/pisa/PIF_124_Figures_Tables.xlsx.

Disaggregating the data by gender, socio-economic status, and immigrant background reveals further insights. Firstly, boys tend to spend more time on digital devices for leisure than girls, which may contribute to differences in academic performance. Secondly, students from lower socio-economic backgrounds often have less access to high-quality digital resources, potentially exacerbating educational inequalities (OECD, 2024[2]). Finally, students with an immigrant background report higher levels of distraction from digital devices compared to their native peers.

Thus, in line with PISA 2022 data, the OECD team highlighted that it is important to secure a balance as digital tools are introduced in education. While they offer substantial opportunities for enhancing learning experiences, there is a critical need for policies that mitigate potential distractions and promote effective use. This includes implementing guidelines for responsible digital device usage, providing high-quality digital content and materials and ensuring equitable access to technology across different socio-economic groups. Additionally, equipping educators with the necessary skills to integrate technology effectively into their teaching practices is crucial for maximising the benefits of education technologies.

The OECD team highlighted that it is important to secure a balance as digital tools are introduced in education. While they offer substantial opportunities for enhancing learning experiences, there is a critical need for policies that mitigate potential distractions and promote effective use.

The OECD team also gave an overview of **Ireland**'s PISA 2022 results, highlighting how they provide relevant insights into the opportunities and challenges associated with the use of digital technologies in education. **Ireland**'s academic performance in PISA 2022 was above average, with students achieving high reading scores compared to both EU and OECD countries. Irish students also performed significantly above average in science and mathematics. Although **Ireland**'s performance was above the OECD average, participants in the event pointed out that challenges remain in integrating digital technologies in Irish education.

In the context of the COVID-19 pandemic and its related shift to remote learning, Irish students reported high levels of confidence in using digital platforms for remote learning – e.g., navigating learning management systems and finding online resources – which reflects their adaptability during this period (OECD, 2023[3]). However, they were less confident about managing their own motivation and focus without external prompts. According to PISA 2022, nearly one in three 15-year-olds across OECD countries are in classes where students get distracted by using digital devices during mathematics lessons. In Ireland, 20% of 15-year-olds are in classes where students are distracted by their own device use, and 16% where they are distracted by others' use. These figures are below the OECD averages (30% and 25% respectively) (OECD, 2023[4]). This trend reflects a broader need to develop strategies that promote self-directed learning skills alongside technological competence (OECD, 2023[3]; OECD, 2023[4]). In addition, during this time, disparities in access to quality educational materials and digital devices between socio-economically advantaged and disadvantaged schools persisted. Addressing these inequities is crucial to ensure all students benefit equally from digital tools (OECD, 2023[4]).

Further, during the initial discussion, Teacher Professional Learning emerged as a critical factor in **Ireland**'s efforts to integrate digital technologies in education. In this regard, participants in the event stressed that equipping teachers with the skills to effectively use and teach through digital technologies is essential for successful integration into classrooms. In particular, the participants from Ireland reported that significant progress has been made under initiatives such as **Ireland**'s Digital Strategy for Schools to 2027 and its associated Implementation Plan, which includes substantial investments in infrastructure and

professional development (Department of Education and Youth, 2022[5]). However, some challenges – e.g., addressing digital skills and literacy in the curriculum, managing distractions, ensuring equitable access to resources and fostering self-directed learning – remain. Finally, the discussion highlighted that addressing these challenges through targeted policies and investments will be crucial for optimising the potential of digital tools to support student achievement.

Box 1. Balancing benefits and distractions, the dual impact of digital technologies on student learning

Digital education technologies impact student learning in several areas. Firstly, they can enhance academic performance when integrated into the curriculum with clear educational objectives. Further, students using digital devices for learning can access a wide range of resources, fostering engagement and potentially improving outcomes. However, nearly one in three 15-year-olds are in classes where students get distracted by using digital devices in OECD countries, and the use of digital devices for leisure, especially during school hours, often correlates with distractions and lower academic performance. This is particularly true when students engage in non-educational activities such as social media browsing or gaming. Finally, digital education technologies influence classroom dynamics and student well-being. They provide opportunities for interactive learning and collaboration but also pose challenges like increased distraction and anxiety related to device dependency. Effective management of digital device use in educational settings is crucial to maximise their benefits while minimising potential drawbacks.

Source: OECD (2024), "Managing screen time: How to protect and equip students against distraction", *PISA in Focus*, No. 124, OECD Publishing, Paris, https://doi.org/10.1787/7c225af4-en.

2 Measuring the impact of digital technologies in education

After the introductory session, the second part of the event focused on the measurement of the impact of digital technologies in education. The complexity of this issue was a key point of discussion during the event, as participants highlighted that assessing the impact of digital technologies in education is a complex and ongoing process. As stated by several authors (Golden, 2020[6]; Hallie Preskill et al., 2013[7]), policy evaluation in education entails understanding complex contexts and relationships within education systems, as well as the specific reforms being implemented. Part of this involves the use of indicator-based frameworks, which allow policymakers to assess the availability, use, and outcomes of, in this case, digital technologies in schools. These frameworks track metrics such as ICT infrastructure (e.g., device-to-student ratios and internet access), usage patterns in classrooms, and learning outcomes like academic performance and digital skills.

For instance, international testing frameworks like PISA offer data and overviews of students' ability to solve complex problems, think critically, and communicate effectively. The same applies for the testing framework of ICILS (International Computer and Information Literacy Study) in relation to students' computer and information literacy (CIL) and computational thinking (CT)[2]. However, while these indicators provide valuable insights into usage patterns and correlations, they do not establish causality or fully capture the nuanced effects of digital technologies on educational outcomes (see Box 1). Because of this, some authors (Golden, 2020[6]; Hallie Preskill et al., 2013[7]; Daniel L. Stufflebeam, 2001[8]) stress the need for more comprehensive evaluation strategies that integrate both quantitative and qualitative methods to fully understand the complexities of educational reforms.

Qualitative and mixed methods assessments – such as interviews or focus groups with teachers, students, and parents – provide crucial context for understanding how digital technologies are perceived and used in practice. Surveys and classroom observations, for example, can complement self-assessment data by offering additional insights into how digital technologies are used and how their use correlates with student engagement and teacher practices (Joint Research Centre, 2023[9]). These methods allow policymakers to consider stakeholder needs, identify implementation barriers, and, ultimately, find ways to address them. Though these data sources can help policymakers track trends and correlations, to assess actual impact and causality mechanisms, additional research approaches – such as experimental designs (e.g., Randomised Control Trials) and longitudinal studies – are required.

Randomised Control Trials (RCTs) allow policymakers to establish causal links between technology use and educational improvements by comparing outcomes between treatment and control groups (Cukurova and Luckin, 2019[10]). Therefore, this type of study offers robust evidence to inform scalable policy decisions and identify effective practices across diverse educational contexts (Golden, 2020[6]). Longitudinal studies allow policymakers to track the long-term effects of digital technologies on students' learning trajectories, career readiness, and overall educational equity. This methodology allows to assess the sustainability of reforms as well as their impact over time (Golden, 2020[6]).

To assess actual impact and causality mechanisms, […] research approaches – such as longitudinal studies or experimental designs (e.g., Randomised Control Trials) – are required.

On the grounds of these strategies and regular evaluations, policymakers can monitor progress and adjust policies based on evidence. By using these methodologies, policymakers can gather robust evidence to support informed decision-making, enabling governments to design policies that maximise the benefits of digital tools while addressing challenges such as inequity and misuse. This evidence-based approach ensures that investments in educational technology contribute meaningfully to achieving broader educational goals. To achieve this, comprehensive digital education strategies are essential in addition to these methods. In fact, strategies can guide policy efforts, as they set clear objectives for the integration of technologies with measurable targets (e.g., for infrastructure development, teacher education, professional learning, and student outcomes) (OECD, 2023[11]).

The experiences of Estonia, France, Sweden and Ireland

Estonia

Estonia[3] is widely acknowledged as a pioneer in the integration of digital technologies into education, serving as a model for other countries. In 1996, **Estonia** launched the Tiger Leap programme (Tiigrihüpe), which was critical for the country's digital policy (Tiger Leap Foundation, 2007[12]). This ambitious initiative aimed to digitise schools by providing them with robust technological infrastructure, promoting digital literacy, and enhancing teachers' ability to effectively integrate technology into their teaching. The goal was to ensure that technology would support the achievement of learning outcomes as outlined in the national curriculum. By the early 2000s, all Estonian schools had internet access and computer labs, creating a strong foundation for the subsequent integration of digital tools (Digital Skills & Jobs Platform, 2022[13]). Building on the Tiger Leap programme, Estonia will launch the AI Leap 2025 (TI-Hüpe 2025)[4] programme to provide students and teachers with free access to leading AI tools as well as equip them with the skills needed to use them effectively in education (e-Estonia, 2025[14]).

Further, since 2014, **Estonia**'s national curriculum has included digital competence as a core objective, further embedding technology into education. This focus has been reinforced by frameworks such as the Lifelong Learning Strategy 2020 and the ongoing Education 2035 strategy, which stresses the role of technology in improving the quality and inclusiveness of education (Figure 2). Additionally, according to the expert from **Estonia**, the country has made efforts to combine its infrastructure investments with initiatives to boost digital skills. For instance, the ProgeTiger programme, launched in 2012, promotes coding, robotics, and other technological competencies among students and educators. The programme also supports continuous teacher training to ensure educators can effectively integrate digital tools into their teaching practices (Education Estonia, 2020[15]).

Despite these advancements, **Estonia** faces challenges in measuring the direct impact of digital technologies on educational outcomes. As current metrics primarily capture access to digital technologies and digital competencies, there is a recognised need for longitudinal research and comprehensive metrics that link digital device usage to academic performance and critical thinking development (Education Estonia, 2024[16]).

Figure 2. Underlying principles and values of Estonia's Education 2035 Strategy

Source: Ministry of Education and Research (Estonia) (2021[17]), *Education Strategy 2021–2035*, https://www.hm.ee/sites/default/files/documents/2022-10/haridusvaldkonna_arengukava_2035_kinnittaud_vv_eng_0.pdf

To address this gap, **Estonia** has initiated studies such as the DigiEfekt Study (2020–2023). Conducted by the University of Tartu, this research examines how digital technologies influence student outcomes in mathematics, science, and digital literacy (Education Estonia, 2024[16]). Early findings suggest that lessons leveraging interactive digital tools to connect prior knowledge with new concepts yield positive outcomes, whereas passive substitution of traditional tools with digital ones has limited effects (Education Estonia, 2020[15]). Crucially, this study found that the nature of technology use is more influential than frequency alone. Interactive or constructive uses of technology demonstrated positive effects on learning outcomes, while passive uses resulted in no or even negative effects depending on the specific learning outcomes measured. These findings underscore the importance of considering not just how often, but how effectively technology is used in educational settings to maximise its positive impact on student learning. Future research and assessments may, thus, need to focus more explicitly on the quality and effectiveness of technology integration to provide a more comprehensive picture of its impact on education.

France

In his intervention, the expert from **France**[5] highlighted that his country has adopted a comprehensive, data-driven approach to integrating digital technologies in education. According to the expert, central to this strategy is EduPilote, a platform aiming to collect, aggregate, and analyse data on the use of digital tools and resources across public, lower secondary (11 to 15 years old) and upper secondary (15 to 18 years old) schools nationwide. As of April 2025, a functional prototype of the platform is being used at the Direction du Numérique pour l'Éducation (DNE)[6]. A first version shall be released to the public by the end of 2025. By leveraging features such as dynamic navigation and data visualisation tools, EduPilote will enable policymakers, local authorities, and educators to track and evaluate the impact of digital technologies on teaching and learning outcomes. This platform will, hence, provide detailed insights into device usage and engagement levels, facilitating informed decision-making tailored to the specific needs of individual schools and districts.

Complementary to this upcoming platform, **France**'s "Territoires Numériques Éducatifs" (TNE) programme supports the pedagogical use of digital resources across 12 selected regions through four main goals: 1) train teachers; 2) equip classrooms; 3) make digital resources and tools available; and 4) support parents. The programme offers webinars, training opportunities, digital resources across a variety of digital related topics, as well as a variety of "educational digital solutions" (e.g., online libraries, learning language tools, etc.) for all educational levels (Ministère de l'Éducation nationale, 2024[18]).

The French expert also mentioned that the 2015 "Digital Plan for Education" was pivotal in supporting the adoption of digital technologies at every educational level in **France**. According to the expert, this plan prioritised teacher training, the development of high-quality digital content, and the integration of technology into curricula to enhance teaching in primary and lower secondary schools. Complementing this effort, the ongoing "École Numérique Rurale (ENR)" programme provides funding for schools to acquire digital tools and resources. This programme has played a critical role in addressing disparities by ensuring equitable access to technology across both urban and rural areas (Ministère de l'Éducation nationale et de la Jeunesse, 2024[19]).

To assess the effects of the 2015 Digital Plan for Education, the Direction de l'Évaluation de la Prospective et de la performance (DEPP) conducted the Élaine (i.e. the Longitudinal Evaluation of Activities related to Digital Education) study in 2017. The study tracked over 5,000 students in 7th and 8th grade and focused on the Plan's deployment of mobile digital equipment in classrooms. Specifically, it assessed two initiatives: the "1-to-1 Tablets" initiative (for both in-school and at-home use) and the "Mobile Classrooms" (MC) initiative (for individual and shared use in the classroom only). Findings showed increased tablet access and usage in schools, alongside improvements in students' digital, mathematical, and language (reading and oral comprehension) skills. The 1-to-1 Tablets programme yielded stronger educational benefits than the MC initiative. However, impacts varied by subject, grade level, and were most significant among students from low socio-economic status and in disadvantaged schools (Direction de L'Évaluation de la Prospective et de la performance (DEPP), 2017[20]).

Currently, **France**'s "Digital Strategy for Education 2023–2027" – i.e. the official digital strategy developed by the then ministère de l'Éducation nationale et de la Jeunesse (now ministère de l'Éducation nationale, de l'Enseignement supérieur et de la Recherche) – provides a roadmap for integrating digital technology into the French education system for the 2023 to 2027 period. The strategy outlines the vision, priorities, and key actions set out to support digital transformation in French schools and to ensure that students and educators develop the digital skills needed in today's world. Further, the strategy relies on four main axes: 1. A committed ecosystem in support of a shared public policy; 2. Teaching digital literacy to build citizenship and digital competence; 3. An educational community supported by a thoughtful, sustainable, and inclusive digital offering; 4. A renewed framework for a (ministerial) information system designed to serve its users (Ministère de l'Éducation nationale et de la Jeunesse, 2023[21]).

Despite significant progress in **France**, challenges remain. Disparities in digital literacy levels among teachers and students persist, prompting targeted professional development initiatives and resource-sharing platforms to bridge these gaps. Additionally, systemic issues such as inconsistent internet connectivity in rural areas and growing cybersecurity risks continue to hinder full-scale implementation. Programmes like "Numérique Inclusif" aim to address these inequities by promoting broader access to digital tools and skills (Ministère de l'Éducation nationale et de la Jeunesse, 2024[19]).

Sweden

Sweden[7] has long been recognised for its high level of digital inclusion in education, supported by widespread access to devices and robust digital infrastructure. In his intervention, the expert from **Sweden** brought forward that schools in the country began introducing computers into classrooms as early as the 1980s, and that authorities (local government or municipalities) have consistently invested in digital tools and platforms to support learning. Since 2017, the National Digitalisation Strategy for the School System

shapes the integration of digital resources in education with the aim of enhancing teaching quality and equipping students with critical digital competencies. The strategy focuses on three key areas: ensuring equal access to digital tools, providing adequate infrastructure, and promoting research on digitalisation's impact (Andric, 2023[22]).

One of the key characteristics of **Sweden's** digital education is its effort to emphasise critical thinking and media literacy. Recognising the prevalence of misinformation online, Swedish schools have incorporated media literacy into their curricula since 2018. This includes teaching students how to critically evaluate digital content and understand programming concepts in technology subjects. These efforts align with **Sweden**'s broader goal of preparing students to navigate an increasingly complex digital world (European Commission, 2024[23]).

To build on its successes, **Sweden** is exploring ways to enhance data collection and analysis related to digital education. Initiatives aimed at linking device usage with academic performance metrics have been piloted under the National Digitalisation Strategy. These efforts seek to develop a more comprehensive understanding of how technology impacts classroom learning outcomes (Skolverket, 2022[24]). This experience underscores the importance of balancing innovation with rigorous evaluation to ensure that digital tools contribute meaningfully to student learning.

Despite these efforts, **Sweden** has faced challenges in the systematically measuring the direct impact of digital technologies on learning outcomes. For example, it has been difficult to examine how digital competencies – assessed by studies as ICILS – translate to improved results in subjects like mathematics or science (Fraillon et al., 2020[25]). Qualitative evidence from educators suggests that digital tools promote student engagement, creativity, and collaboration. According to the expert, many teachers report using digital technologies to facilitate group projects, enable real-time feedback, and customise lessons to meet individual learning needs. However, systematic data collection linking these practices to measurable improvements in academic outcomes remains limited. This gap is also highlighted in reports published by the Swedish National Agency for Education (Skolverket). The reports also emphasise the need for enhanced data collection mechanisms (Skolverket, 2022[24]).

According to the expert, **Sweden** also faces equity challenges in its digital education landscape. While most students have access to devices at school, disparities in digital literacy and home internet access persist, particularly among socio-economically disadvantaged groups. To address these gaps, the government has launched initiatives such as the provision of funding for schools to purchase devices and implement teacher training programmes on effective digital pedagogy. Additionally, the "Skoldigistöd" initiative organised by the SALAR (Swedish Association of Local Authorities and Regions) supports equitable access to technology by helping schools procure and implement digital tools effectively (OECD, 2023, p. 299[26]).

Based on the interventions from the different experts, several challenges are common when it comes to measuring the impact of digital technologies in education across **Estonia**, **France**, and **Sweden**. One significant issue is the lack of standardised metrics. There is a notable gap in measurements that go beyond access to digital technologies. While international assessments like ICILS (in which **Ireland** has not participated to date) measure digital competencies, there is a lack of comprehensive metrics assessing how the use of technology directly influences academic performance, critical thinking, and other learning outcomes, as discussed in the relevant section of this document (OECD, 2023[27]). These countries also lack comprehensive longitudinal studies that directly link digital device usage to learning outcomes, something that is missing in many other OECD countries as well (OECD, 2023[27]).

Ireland

In their turn, representatives from **Ireland**[8] offered a presentation of their own situation. **Ireland's** approach to measuring the impact of digital technologies in education reflects a structured and stakeholder-driven

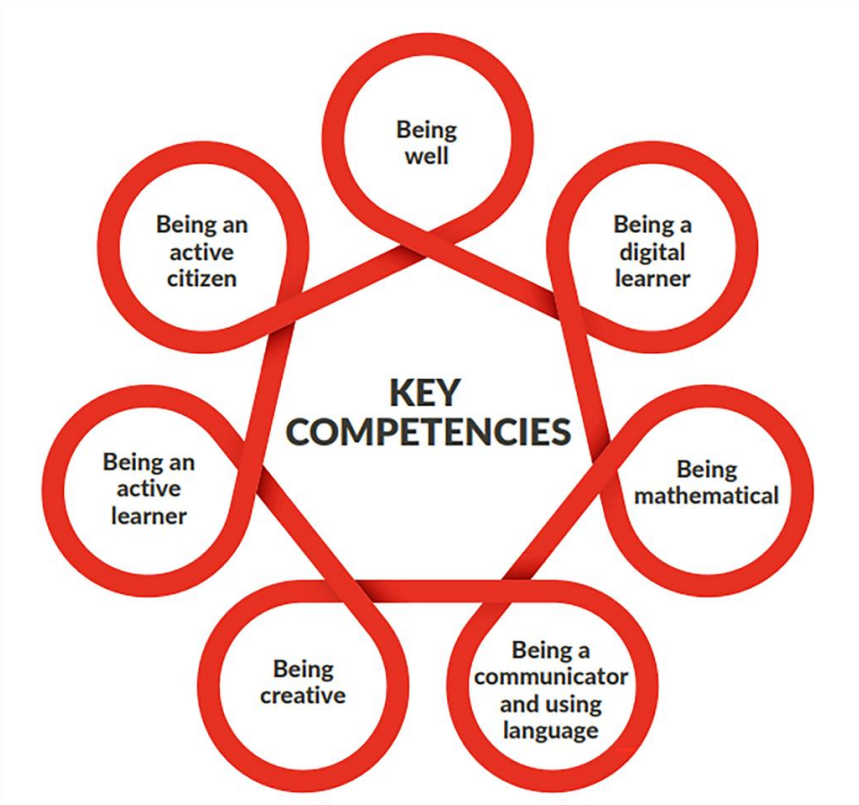
strategy, with room to enhance its evaluation mechanisms. A key element for this is the Digital Learning Framework (DLF), a tool which assists schools and teachers to plan and monitor their progress in embedding digital technologies into teaching and learning. The tool provides schools with clear descriptors of relevant digital competencies for teachers and school leaders, emphasising the use of technology to improve teaching, learning, and assessment outcomes. The DLF also provides a Digital Learning Planning Cycle teachers and schools can engage with as develop their own digital plan. The Cycle is composed of six steps (see Figure 3): the first three (i.e. "identify", "gather evidence", and "analyse and make judgements") form the investigation phase, the fourth ("write and share reports") represent the creation phase, the fifth (i.e. "put improvement and plan into action) represents the implementation phase, and the last ("monitor actions and evaluate impact") represents the evaluations stage (Oide Technology in Education, 2024[28]). By focusing on the alignment of digital practices with pedagogical goals, the DLF ensures that digital tools contribute meaningfully to student achievement.

Figure 3. The six steps of the Digital Learning Planning Cycle

Source: Oide Technology in Education (2024[29]), *Digital Learning Framework - Planning Resources*, https://www.dlplanning.ie/

Further, the **Irish** Primary Curriculum Framework outlines "being a digital learner" as one of seven key competencies to embed across all curriculum areas and subjects (see Figure 4). This aims to make children curious, creative, confident, and critical users of digital technology while also fostering a responsible, safe and ethical use of technology. The framework also proposes that digital technology use would become part of the curriculum area of Science, Technology, and Mathematics with its own subject specification for third to sixth class. This approach highlights **Ireland's** aim to integrate digital competencies into everyday teaching practices across different educational levels, especially with strategies aligned to the technological advancements shaping the future.

Figure 4. The Primary Curriculum Framework's core competencies



Source: National Council for Curriculum and Assessment (2023[30]). *Primary Curriculum Framework for Primary and Special Schools*. https://curriculumonline.ie/getmedia/84747851-0581-431b-b4d7-dc6ee850883e/2023-Primary-Framework-ENG-screen.pdf

At the post-primary level (i.e. junior and senior cycle levels), students' competencies are further built on this foundation. The junior cycle English curriculum supports students in using literacy skills to manage information, and to find, evaluate, synthesise, and communicate information across various media. The curriculum emphasises the use of non-literary texts – such as advertisements, media broadcasts, and opinion pieces (e.g., articles, speeches, social media posts) – to help students interpret meaning, assess effectiveness, and respond critically to digital media. The aim is for students to analyse key ideas, style, tone, content, and overall impact of these texts in a systematic way.

At the senior cycle, the Key Competencies Framework informs the development of subjects, modules, and programmes. **Ireland**'s National Council for Curriculum and Assessment (NCCA) has published a schedule for the redevelopment of all Senior Cycle subjects. The Framework emphasises the interconnectedness of key competencies, literacy, and numeracy, and how the development of one supports the others. This framework also underscores how key competencies are enhanced when students use digital tools to aid their learning. Additionally, in line with the Digital Strategy for Schools (DFL), there are ongoing efforts to explore how the use of digital technologies can be incorporated into the curriculum and assessment of revised Leaving Certificate subjects, as appropriate. Moreover, a revised specification for the Leaving Certificate Computer Science will be introduced in September 2027 as part of the third phase of subject revisions. The development group for this revision convened on September 2024 under NCCA structures, drawing on a brief and research from other jurisdictions to inform its work.

Finally, online safety remains a central focus of **Ireland**'s strategy to ensure that the benefits of digital tools are not undermined by risks like cyberbullying or distractions. To this purpose, **Ireland** has developed resources like the Webwise programme, which supports schools in promoting a safe and responsible use

of digital technologies (Department of Education and Youth, 2024[31]). Further, in terms of funding and infrastructure, **Ireland**'s ICT Grant Scheme and the Schools Broadband Programme support schools in acquiring the necessary resources to enhance digital learning environments. These initiatives provide schools with the autonomy to tailor their digital leaning plans while ensuring alignment with national standards and guidelines. Lastly, recent advances in artificial intelligence, particularly generative AI, have prompted discussions about their potential impact on assessment. The State Examinations Commission (SEC) has been tasked with commissioning research on the role of generative AI in teacher-based assessment, exploring both the opportunities and challenges these technologies present.

Key considerations for Ireland

At the end of this session, several key considerations emerged. To begin with, **Ireland** has made notable progress in supporting the embedding of digital technologies in education. This was achieved through initiatives as the Digital Strategy for Schools, the Digital Learning Framework (DLF), the curricular reforms, and the provision of a wide range of professional learning opportunities for teachers which all support a structured and pedagogical approach to the integration of technology into teaching and learning. Additionally, the ICT Grant Scheme and Schools Broadband Programme have bolstered infrastructure, supporting schools in enhancing their digital learning environments. However, **Ireland** could learn from the experiences of **Estonia**, **France**, and **Sweden** on how to address some of its measurement gaps.

Estonia's concrete effort to provide rigorous data on the effects of digital technologies on student learning, could offer insights that **Ireland** might adopt to strengthen its long-term monitoring of digital tools' effectiveness. **France**'s EduPilote platform (prototype) exemplifies a data-driven initiative that links technology use to learning outcomes, a strategy **Ireland** could explore to improve evidence-based policy decisions. On this, **Sweden** has also identified the need to address the challenge of linking digital tools usage directly to academic performance; a challenge that can be tackled by developing more robust, standardised metrics and experimental studies (like RCTs) to assess the full impact of digital technologies. By building on these international experiences, **Ireland** could refine its approach to measuring the real-world outcomes of its digital education strategies, ensuring that technology adoption translates into tangible improvements in student outcomes.

3 Ensuring quality, competence and equitable access in digital education

Discussions during the peer-learning event emphasised that achieving equitable and effective digital education systems requires addressing interconnected challenges of infrastructure, teacher preparedness, and resource quality through system level policies. According to the framework of the OECD project *Resourcing School Education: Policies for the Digital Transformation of Education and Future-Readiness of Teachers* (OECD, 2025[32]), considering these elements is fundamental to address barriers such as disparities in access to quality digital infrastructure, lack of necessary skills for the pedagogical use of digital resources in and out of the classroom, assessment of digital resources, and sustainable financing among others. Doing so through a systemic policy approach can ensure coherence across policy areas, hence establishing a comprehensive and integrated policy environment that enables effective uses of digital resources.

As highlighted at the start of the event, teacher professional learning remains a critical enabler of educational success. However, only 43% of educators in OECD countries report confidence in using digital tools pedagogically, despite widespread technical training (OECD, 2023[11]). To address this gap, some accredited professional development programmes have focused on AI-driven lesson differentiation and digital assessment literacy. These programmes demonstrate the value of grounding training in the Technological, Pedagogical, Content Knowledge (TPACK) model, which integrates technological, pedagogical, and subject-specific expertise (Koehler, 2013[33]). This approach aligns with broader OECD recommendations for competency frameworks that prioritise continuous, practice-oriented learning ecosystems – such as peer coaching networks and micro-credentialling systems tied to career progression – over one-off workshops (OECD, 2023[11]).

Further, fostering student self-direction requires deliberate policy architectures rather than assuming innate digital competence. To this end, systemic models that embed self-regulated learning objectives into curriculum standards show promise, particularly when supported by learning analytics dashboards that provide real-time feedback. Evaluations of such approaches point to measurable improvements in problem-solving competencies (OECD, 2023[27]). Nonetheless, the OECD stresses the need for parallel investments in teacher training to interpret analytics effectively (2023[27]).

Underpinning all these elements are governance frameworks that coordinate cross-sector priorities. An example of the needed systemic coordination would be coupling inter-ministerial collaborations that align infrastructure investments and pedagogical goals with procurement standards that ensure edtech accessibility and data ethics. As highlighted in the OECD's *Shaping Digital Education: Enabling Factors for Quality, Equity and Efficiency* report (2023[11]), if monitoring focuses on classroom integration rather than only on device distribution metrics, such architectures might transform the usage of digital tools from isolated interventions into drivers of pedagogical renewal.

The experiences of France, Sweden, Estonia and Ireland

France

In terms of quality, competence, and equitable access in digital education, **France** has adopted a centralised and structured approach to integrating digital technologies in education, aiming to address quality, competence, and equitable access. The Pix platform plays a central role in this, by offering tailored, skill-oriented assessments that promote high-quality digital literacy across all educational levels. With a comprehensive curriculum mapped to real-world applications, Pix ensures that students and educators develop both technical and critical thinking skills, essential for navigating a rapidly evolving digital landscape. The platform is a key driver of competence development, as it enables educators to evaluate and improve their own digital skills through self-paced learning modules.

To cater to the specific needs of different user groups, Pix offers two distinct versions: one for students and another one for teachers. The student version, simply called Pix, is designed to assess, develop, and certify digital skills throughout their education, from lower secondary to upper secondary schools. It includes mandatory certifications at key stages and helps prepare students for the digital demands of further education and the workforce. For teachers and education staff, a specialised version called Pix+ Édu has been developed. This version focuses on enhancing educators' professional digital competencies, aligning with the European DigCompEdu framework and the French Cadre de Référence des Compétences Numériques pour l'Éducation (CRCN-Édu). Pix+ Édu offers self-assessment, thematic training pathways, and the opportunity for teachers to validate their digital skills through certification. This dual approach ensures that both students and educators are equipped with the necessary digital competencies to thrive in the modern educational environment.

The expert from **France** also mentioned that to address equity, support for the provision of digital devices is provided to schools, especially to rural and economically disadvantaged ones. For instance, the 2015 Digital School Plan provided schools with financial support for purchasing devices, training teachers, and modernising IT infrastructure. Further, since 1981, the Territories of Educational Priority (REP) initiative foresees additional funding and training to ensure students located in disadvantaged areas have equal opportunities to access digital resources (Éduscol, 2024[34]). Finally, the expert referred once again to the EduPilote platform prototype (also mentioned in section Measuring the impact of digital technologies in education), which monitors the usage and impact of digital tools in schools. By collecting real-time data on digital education practices, the platform allows policymakers to identify disparities and inefficiencies, enabling adaptive and evidence-based interventions. However, despite these measures, challenges persist in fully bridging the rural-urban divide and ensuring universal access to high-speed broadband, which remains uneven in some regions (Ministère de l'Éducation nationale et de la Jeunesse, 2024[19]).

Sweden

In contrast, the expert from **Sweden** explained how their decentralised model reflects the country's broader commitment to autonomy and innovation and objective of ensuring quality, competence, and equitable access by empowering local stakeholders. Municipalities and schools have the flexibility to design tailored digital education programmes, ensuring that strategies are locally relevant and responsive to specific needs. This decentralisation aims to give more autonomy to local agents, so that schools may experiment with innovative approaches (e.g., gamified learning, digital storytelling, or virtual reality-based teaching).

Accordingly, to promote teacher competence equitably, **Sweden** provides professional development programmes funded through municipal budgets and supported by national guidelines. Initiatives like the National Digitalisation Strategy for the School System outline competency goals for teachers and encourage the integration of digital tools into core subjects. However, the decentralised system has resulted in disparities in training quality and access. For instance, rural municipalities often struggle to

allocate resources for advanced teacher training or state-of-the-art digital tools, creating inconsistencies in digital education quality across the country.

In this perspective, equitable access remains a significant challenge in **Sweden**. Schools in wealthier municipalities benefit from better funding, leading to more advanced infrastructure and technology adoption compared to less affluent areas. Efforts to address these disparities without targeted national subsidies for schools in underserved regions is difficult due to the lack of uniform standards, which can hinder broader equity goals. **Sweden**'s experience thus highlights the trade-offs of a decentralised approach: while it fosters innovation, it requires robust mechanisms to ensure consistency and fairness (Skolverket, 2022[24]).

Estonia

The expert from **Estonia** explained that the country's holistic and technology-driven approach to digital education is designed to comprehensively tackle quality, competence and equitable access. Central to **Estonia**'s strategy is the Digital Competency Framework (already mentioned in section "Measuring the impact of digital technologies in education"), which ensures a consistent and high-quality digital education experience across the country by outlining clear standards for students and teachers. This framework stresses critical digital skills, including cybersecurity, data literacy, and creative problem-solving, ensuring that students are not just passive users of technology but active and informed participants in the digital economy.

To address equity issues, **Estonia** has implemented several nationwide initiatives, such as the ProgeTiger programme (also mentioned in section "Measuring the impact of digital technologies in education") which provides free coding and robotics education for all students, regardless of their socio-economic background. The Estonian government also prioritises rural and disadvantaged communities through targeted funding for devices, infrastructure, and teacher training. **Estonia**'s eKool platform exemplifies the country's commitment to equity by providing a free, universal digital interface for students, teachers, and parents, ensuring that all stakeholders have access to real-time information about learning progress.

Despite these achievements, **Estonia** continues to face challenges in ensuring universal access to high-quality broadband in rural areas, as well as in addressing socio-economic disparities in device ownership. Nevertheless, its comprehensive, forward-thinking policies and commitment to data-driven decision-making position it as a global reference in digital education innovation.

> Achieving equitable and effective digital education systems requires addressing interconnected challenges of infrastructure, teacher preparedness, and resource quality through systemic policy design.

Ireland

During this session, the Irish expert highlighted that **Ireland** has taken important steps to address issues of equity, competence, and quality in digital education, while continuing to work on bridging gaps and improving outcomes. The Digital Strategy for Schools to 2027 emphasises inclusion as a key priority. Programmes like Delivering Equality of Opportunity in Schools (DEIS) provide additional funding and resources to schools who have high concentrations of students from socio-economically disadvantaged areas, helping to close the digital divide. DEIS schools and special schools/special classes in schools also benefit from supplementary funding under the ICT Grant Scheme.

In relation to digital competencies, teachers' digital skills have been a key focus, with Initial Teacher Education (ITE) programmes incorporating digital skills training since 2020. Additionally, the Oide (the Irish Department of Education's professional development service for teachers) organises professional

development initiatives, ensuring that educators have access to workshops, resources, and school-based supports to enhance their digital teaching practices. This approach aligns with **Ireland**'s Digital Learning Framework (DLF) (already mentioned in section "Measuring the impact of digital technologies in education") which promotes innovative and reflective teaching methods that integrate technology effectively. In more details, the Oide Technology in Education (TiE) division develops and delivers professional learning opportunities such as online and face-to-face courses, workshops, seminars, webinars and whole-school support.

The Oide TiE also provides advice and support to schools on broadband, ICT Infrastructure matters, and Internet safety. In particular, the latter is fostered through the Webwise program, which promotes the autonomous, effective, and safe use of the internet by young people and has a wide range of training materials and resources available to guide teachers, students and parents on online safety. Webwise develops and disseminates resources that help teachers integrate digital citizenship and online safety into teaching and learning in their schools. With a growing repository of over 25,000 resources and 100 good practice videos tailored to different age groups, the platform supports schools and teachers to create safe digital learning environments for students. Moreover, Oide TiE manages Scoilnet – **Ireland**'s official digital education content portal – which offers a wealth of educational materials, including videos, lesson plans, and good practice guides for educators. Equity of access is further enhanced by **Ireland**'s investment in broadband provision under the Schools Broadband Programme which provides high-speed connectivity to all recognised schools.

However, like other countries, **Ireland** still faces challenges in fully bridging the digital divide, particularly in ensuring universal access to high-quality devices and resources across all socio-economic strata. For example, while DEIS schools receive additional support, ensuring consistency in access and outcomes remains an ongoing priority. In this regard, participants highlighted that **Ireland** has demonstrated a commitment to reducing the digital divide and is working to ensure equitable access to digital resources across all schools. As part of the National Recovery and Resilience Plan, the Irish government issued €50 million in funding to support schools and students at risk due to socio-economic factors. This funding supported schools in addressing the digital divide and aimed to ensure that disadvantaged learners have access to the necessary tools for digital learning. The country is also making efforts to improve digital infrastructure, with many schools benefitting from high-quality broadband and filtered content. Schools then have autonomy in their digital learning plan to determine how to deploy digital technologies including devices, in their schools. In doing so, it is important that schools are supported so that all students experience the same level of access and opportunity.

Key considerations for Ireland

In conclusion, **Ireland** has made significant improvements in addressing equity, competence, and quality in the embedding of digital technologies in teaching and learning. Key assets include its focus on equitable access through initiatives like DEIS and the ICT Grant Scheme, which provide targeted support to socio-economically disadvantaged schools. **Ireland**'s commitment to teacher competence is also evident, with a range of opportunities for digital skills training for teachers and ongoing professional development through the Oide TiE platform at all stages throughout their careers. Furthermore, the country's investment in broadband infrastructure means that all recognised schools are provided with high-speed connectivity through the Schools Broadband Programme operated by the Department of Education.

However, there are areas where **Ireland** could benefit from the experiences of other countries. **Sweden**'s decentralised model offers a valuable experience in promoting innovation and tailoring digital education to local needs, although it also highlights the challenge of ensuring consistency across localities. **France**'s structured approach, including the Pix platform and targeted rural initiatives, could help **Ireland** ensure universal access to digital tools and high-speed broadband. **Estonia**'s holistic, data-driven strategy, which

includes a clear Digital Competency Framework and free nationwide coding programmes, exemplifies how comprehensive and inclusive policies can address both equity and quality on a national scale. By considering some of these aspects, **Ireland** could further strengthen its digital education ecosystem and ensure that all students benefit from the opportunities technology offers.

4 Key factors for successful outcomes in digital education

In this section of the event participants were presented with insights from PISA 2022, thematic OECD reports, and other international sources to highlight several key factors that contribute to successful outcomes in digital education. Such factors include equitable access to digital resources, effective integration of technology into teaching practices, teacher training in digital competencies, and the establishment of clear policies to regulate the use of digital devices in schools. Additionally, measuring and evaluating these initiatives is essential to ensure their effectiveness and scalability. Finally, regular oversight and dissemination of best practices across education systems can further support the continuous improvement of digital education. Several of these points were already raised in other sections of the event as there is a natural overlapping between all these topics.

As mentioned in section 3, “Ensuring quality, competence and equitable access in digital education”, equitable access to digital tools is fundamental for ensuring that all students benefit from digital education. PISA 2022 data highlights disparities in the availability of digital devices and educational materials, particularly between socio-economically advantaged and disadvantaged schools, as well as between urban and rural areas. The data finds that wealthier schools are more likely to have access to high-quality digital resources which can enhance learning outcomes. Addressing these inequities is critical, as unequal access risks widening the achievement gap between different student groups. Thus, governments are encouraged to ensure a minimum baseline of digital resources across all schools to prevent such disparities (OECD, 2024[2]).

Participants also revisited the importance of integrating technology into teaching practices in a purposeful and balanced manner. As highlighted in section 1, “Opportunities and challenges of digital technologies in education: OECD and Ireland’s overview”, PISA data indicates that moderate use of digital devices for learning is associated with improved academic performance. Students who used devices moderately scored higher in mathematics compared to those who did not use them at all or used them excessively. However, excessive use of technology for leisure or non-instructional purposes can also negatively impact learning outcomes by reducing focus and engagement (OECD, 2024[1]). This underscores the importance of using technology intentionally to support educational goals rather than as a substitute for traditional teaching methods.

Further, as highlighted along the entire event, quality teacher training is another critical factor for successful digital education. Teachers need both technical and pedagogical skills to effectively incorporate digital tools into their instruction. Countries like **Estonia** have invested significantly in equipping educators with these high-standard competencies, focusing not only on how to use modern technology but also on how to teach through it. Such training ensures that teachers can select appropriate digital resources, guide students in their use, and foster meaningful learning experiences. Additionally, teacher support during periods of remote learning or school closures has been shown to positively influence student outcomes.

Participants also discussed the idea that clear policies guiding the use of digital technologies in schools are essential for mitigating distractions and maximising their benefits. On this, it was noted that PISA data reveals that nearly one in three 15-year-olds are in classes where students get distracted by using digital

devices, which can significantly lower academic performance (OECD, 2024[1]). Schools that establish clear policies on when and how devices can be used – such as banning mobile phones during lessons or creating technology-free zones – are better able to control these distractions and avoid negative consequences. Moreover, fostering responsible internet behaviour through specific programmes can help students develop critical skills for navigating the digital world.

In sum, successful outcomes in digital education depend on a combination of equitable access to resources, thoughtful integration of technology into pedagogy, comprehensive teacher training and robust policies governing device use. These measures ensure that technology serves as a tool for enhancing learning rather than a source of inequality or distraction.

The experiences of Estonia, France, Sweden and Ireland

Previous sections of this summary have already addressed, either explicitly or implicitly, important factors for promoting the responsible use of digital education. To avoid repetition, this section will be shorter than the previous ones and will highlight one key success factor from each country as an illustrative example. These include comprehensive community consultation in **Estonia**, strong stakeholder engagement in **France**, purposeful curriculum integration of digital competencies within broader learning objectives in **Sweden**, and policy frameworks that balance innovation with inclusivity in **Ireland**.

Community consultation in Estonia

In his intervention, the representative from **Estonia** explained how community consultation plays a critical role in shaping digital education strategies. The involvement of parents, teachers, and school leaders in the development of digital education policies ensures that these strategies are tailored to local needs and contexts. This collaborative approach not only fosters a sense of ownership among the community but also supports the development of policies that are both effective and sustainable (Education Estonia, 2024[16]).

Stakeholder consultation in France

According to the expert from **France**, the country's approach to fostering digital citizenship also exemplifies the importance of engaging stakeholders. The French government has made significant legislative efforts to raise awareness about responsible digital use, targeting both parents and students. These initiatives aim to educate the wider community on issues such as online safety, digital etiquette, and the ethical implications of digital technologies. By including parents in the conversation, **France** attempts to create a cohesive environment where students, educators, and families work together to promote responsible digital practices (Ministère de l'Éducation nationale et de la Jeunesse, 2024[19]).

Integrating digital competencies within the curriculum in Sweden

Sweden's education system provides valuable insights, particularly through its purposeful integration of digital competencies within broader learning objectives in their curriculum. The Swedish curriculum is designed to ensure that students not only acquire digital skills but also understand the ethical implications of their digital interactions. According to the expert from **Sweden**, the country's holistic approach highlights the importance of integrating digital literacy into general education, making it a natural part of students' broader learning experiences. By aligning digital competencies with wider educational goals, **Sweden** promotes a responsible, balanced approach to digital technology in schools (Skolverket, 2022[24]).

Successful outcomes in digital education depend on a combination of equitable access to resources, thoughtful integration of technology into pedagogy, comprehensive teacher training and robust policies governing device use.

Together, these examples underscore the need for balanced policies that encourage innovation while safeguarding inclusivity and responsible usage. Participants in the event in Dublin agreed that the key to achieving sustainable outcomes is the active involvement of all stakeholders, including policymakers, educators, parents, and students. Successful initiatives require collaboration across multiple sectors and levels of society, ensuring that digital education strategies are not only forward-thinking but also equitable and accountable.

Balanced policy frameworks in Ireland

In a dedicated presentation, **Ireland** explained how the country has embraced several strategies and goals to ensure successful outcomes in digital education. For example, their Digital Strategy for Schools to 2027, that was also referred to in other sections of the event given its multiple aims, is underpinned by three pillars: 1) supporting the embedding of digital technologies in teaching, learning and assessment; 2) digital technology infrastructure; and 3) looking to the future: policy, research and digital leadership. To ensure that all voices were given the opportunity to contribute and inform its development, the strategy was informed by a wide and extensive consultation process. This included management bodies of the schools, teacher unions, students, parents, industry and non-governmental organisations including disability groups. This comprehensive approach aimed to create a balanced, inclusive, and forward-thinking education system.

Other complementary strategies include **Ireland**'s Literacy, Numeracy and Digital Literacy Strategy 2024-2033: Every Learner from Birth to Young Adulthood and its implementation plan to 2028 which covers all learners from early years, primary and post primary levels. The Strategy addresses a variety of themes from inclusive learning to professional development and stakeholder engagement (see Figure 5) with the goal of ensuring that all young people leave school with the digital skills required for everyday life and further studies. Additionally, the STEM Education Policy Statement 2017–2026 aims to nurture Science, Technology, Engineering and Mathematics (STEM) in Irish learners from a young age, to ensure they develop skills such as curiosity, inquiry, problem-solving, creativity, ethical behaviour and persistence to operate in an increasingly digital world.

Like in the case of the other countries participating to the event, **Ireland** considers that Initial Teacher Education (ITE) and Continuous Teacher Professional Development (CPD) are vital to ensure the success of digital education initiatives. In **Ireland**, ITE programmes now include digital skills as a core element of all programmes for pre-service teachers[9]. Further, ongoing professional development is provided by Oide, which offers workshops, resources, and on-site support for schools. The expert from **Ireland** also stressed that teacher training programmes should be designed to equip teachers with the skills to integrate digital tools into the curriculum effectively and that ongoing support and resources must be provided to ensure teachers are able to stay up to date with new technologies and pedagogical practices. The development of a national framework for teacher professional learning in digital literacy (e.g., **Estonia**'s Digital Competency Framework) could guide these efforts and ensure alignment with curriculum reforms.

Figure 5. Key themes of the Literacy, Numeracy and Digital Literacy Strategy 2024-2033

Source: Department of Education and Youth and Department of Children, Equality, Disability, Integration and Youth (2023[35]), *Ireland's Literacy, Numeracy and Digital Literacy Strategy 2024–2033: Every Learner from Birth to Young Adulthood*, https://oide.ie/wp-content/uploads/2023/09/Irelands-Literacy-Numeracy-and-Digital-Literacy-Strategy-2024-2033-Every-Learner-from-Birth-to-Young-Adulthood.pdf

Moreover, **Ireland** recognises the importance of regulating device use to maximise benefits and through resources such as Webwise, mentioned already in multiple occasions, schools promote responsible internet usage and online safety. The expert pointed out that all schools are advised to have an Acceptable Use Policy to address the safe, acceptable and responsible use of the internet and digital technologies within the school. These complementary policies, aimed at reducing distractions, promote responsible digital engagement at a societal level. However, as mentioned before, while **Ireland** excels in providing a robust framework and equitable resource allocation, it probably lacks the granular, real-time data tracking capabilities foreseen with **France**'s EduPilote platform (prototype), which will monitor digital resource effectiveness at the local level. Incorporating such capabilities would allow **Ireland** to dynamically adapt its strategies and ensure more precise interventions.

Finally, for **Ireland**, as discussed in previous sections of the event, one of the key challenges in integrating digital technologies into the classroom is measuring their impact on learning outcomes. Hence, it would be essential to establish a baseline for digital competencies at the start of any initiative, followed by pre- and post-evaluations to track progress. Schools need access to practical tools for measuring not only the effectiveness of digital interventions but also how they contribute to overall learning outcomes.

Key considerations for Ireland

Ireland has made notable progress in promoting successful outcomes in digital education, particularly through its comprehensive Digital Strategy for Schools to 2027 and programmes like DEIS, which ensure equitable access to digital technologies for teaching and learning. The focus on teacher training and CPD equips educators with the competencies needed to integrate technology effectively into teaching. Furthermore, **Ireland**'s emphasis on responsible digital engagement, through resources like Webwise and policies such as the ban on the use of mobile phones for post-primary schools announced in 2024, mirrors international efforts to address online safety and prevent distractions. However, **Ireland** can draw valuable considerations from the experiences of **Estonia**, **France**, and **Sweden**.

Estonia's approach to community consultation to shape digital education strategies could inspire Ireland to enhance its already strong stakeholder engagement, ensuring policies are more closely aligned with local needs. In fact, **Ireland**'s approach to stakeholder engagement is similar to that of **France**, as they are both aimed at creating a cohesive environment where students, educators, and families work together to promote responsible digital practices. Additionally, **Sweden**'s holistic approach to digital literacy, which integrates digital competencies into broader learning objectives, resembles **Ireland**'s perspective of digital education not just as a standalone subject, but as an integral part of overall student development. By reflecting on some of these similarities and differences, **Ireland** can refine its digital education policies, making them more responsive and impactful.

5 Conclusions and main considerations

Discussions at the event highlighted that there is evidence of the positive potential of digital technologies on student learning outcomes, especially when they are used with a clear pedagogical purpose and in moderation (University of Tartu, 2024[36]). Indeed, findings from international assessments, such as PISA and ICILS, indicate that students who engage with digital tools for learning in a balanced way tend to perform better academically. However, this potential is tempered by the risks associated with excessive screen time, particularly when digital technologies are used for non-educational purposes.

Hence, as for other countries, the integration of digital technologies presents both transformative opportunities and significant challenges for the Irish education system. The participants concluded that, while there are no one-size-fits-all solutions, an effective approach is for policymakers to implement guidelines that promote responsible digital device use, while ensuring that these resources are used to enhance educational outcomes. In this regard, the experiences of **Estonia**, **France**, and **Sweden** provide valuable considerations for **Ireland** as the country progresses in its implementation of digital education, particularly for the assessment of its impact on student learning. As mentioned in different parts of this summary, **Ireland**'s Digital Strategy for Schools to 2027 and its associated Implementation Plan reflect the county's recognition of the challenges and opportunities that come with digital education. The strategy emphasises a holistic approach, integrating digital technologies into teaching and learning while addressing disparities in access and competence.

Another key takeaway from the peer-learning event is the importance of ensuring equitable access to digital tools and resources. As demonstrated by the challenges faced in **Estonia**, **France**, and **Sweden**, disparities in access – whether due to socio-economic factors, rural location, or other barriers – can hinder the success of digital education initiatives. **Ireland** must continue its efforts to ensure that all students benefit from digital education technologies and have equitable opportunities to fulfil their potential. Initiatives such as the Delivering Equality of Opportunity in Schools (DEIS) programme and the ICT Grant Scheme showcase **Ireland**'s commitment to reducing the digital divide and ensuring equitable access for all students. Indeed, programmes like the Schools Broadband Programme – which provides high-speed internet connectivity to all schools – are already oriented to address this challenge, though it seems that sustained investment will be necessary to ensure lasting equity. These measures are vital for bridging gaps in access and achieving equitable outcomes.

Teacher competence and continuous professional development also emerged as critical factors in the successful integration of digital technologies in education. **Ireland**'s Digital Learning Framework (DLF) provides a structured approach to equipping teachers with the skills needed to integrate technology meaningfully into classrooms. Professional development opportunities through the Oide TiE further ensure that teachers remain supported as they adapt to technological advancements. Building on these initiatives will help **Ireland** to foster a teaching workforce that is both confident and proficient in using digital tools.

In addition, to effectively measure and improve the impact of digital technologies on education, it is crucial for **Ireland** to establish a robust system of baseline data to support ongoing monitoring, measurement, and evaluation. This should include leveraging international assessments such as PISA but also ICILS, which

provide valuable data on students' digital literacy and computational thinking. By regularly collecting and analysing this data, **Ireland** can ensure that its digital education policies are not only aligned with international best practices but also responsive to emerging needs and challenges. Integrating indicators on digital access, usage patterns, and student outcomes into national assessments could also help policymakers track progress over time and make informed adjustments to **Ireland**'s Digital Strategy for Schools. Finally, the experiences shared by peer countries underscore the importance of stakeholder engagement. **Ireland**'s focus on engaging parents, educators, and students in its digital education strategy aligns with international best practices. Programmes like Webwise, which promote responsible internet behaviour and online safety, exemplify how inclusive strategies can foster a culture of digital citizenship and collaboration.

In sum, while digital technologies hold significant potential to transform education in **Ireland**, this potential can only be realised through careful planning, collaboration, and continuous evaluation. **Ireland** has made progress in developing comprehensive frameworks, improving equitable access, and supporting teacher development, while continuing to address challenges and learn from international practices**.** However, there is room to enhance data-driven decision-making to dynamically assess and refine digital education policies. By adopting a balanced approach that addresses access, teacher competence, responsible usage, stakeholder engagement, measurement, and oversight, **Ireland** can navigate the complexities of digital education and create a more inclusive and effective learning environment for all students.

Key messages: Balancing the risks and opportunities of digital technologies in education

Drawing on insights from an international peer-learning event hosted by the OECD and Ireland's Department of Education, this document summarises key evidence and insights presented by representatives from Estonia, France, and Sweden as well as practical considerations for Irish policymakers. Key issues addressed include the balance between educational benefits and disadvantages, the importance of equitable access to technology, and the need for teacher competence and professional development. This document briefly highlights the practical options discussed during the event for Irish policymakers to foster a continuous reflection about how to enhance its digital education initiatives, ensuring they are effective, inclusive, and aligned with broader educational goals.

Annex A. Agenda of the international peer-learning event

Theme: Impact of digital devices and education technologies on student learning
Date: 25 November 2024
Location: Dublin, Ireland
Organisers: Ireland's Department of Education and OECD
Format: In-Person Event

Time	Agenda Item	Details
9:30 AM – 10:00 AM	Opening Remarks and Event Objectives	**Welcome Address:** Representative from Irish Department of Education. **OECD Introduction and Objectives of the Day:** OECD Representative
10:00 AM – 10:10 AM	Overview of Irish Experience with Digital Devices in Education	Presentation by Irish Department of Education on the national experience and key insights.
10:10 AM – 11:25 AM	Session 1: Measuring the Impact of Digital Devices in Education	**Presentations by** Estonia, Sweden, and France (10-15 minutes each), followed by Q&A.
11:25 AM – 11:40 AM	Coffee Break	
11:40 AM – 12:55 PM	Session 2: Ensuring Quality, Competence, and Equitable Access	**Presentations by** Sweden, France, and Estonia (10-15 minutes each), followed by Q&A.
12:55 PM – 2:00 PM	Lunch Break	
2:00 PM – 3:15 PM	Session 3: Key Factors for Successful Outcomes in Digital Education	**Presentations by** France, Estonia, and Sweden (10-15 minutes each), followed by Q&A.
3:15 PM – 3:30 PM	Irish School's Experience	**Presentation by Oide TiE**. Oide Technology in Education (Oide TiE) supports Irish schools in integrating digital technologies into teaching, learning, and assessment through professional development, resources, and strategic guidance.
3:30 PM – 4:45 PM	Panel Discussion	**Informal open panel discussion** with the three countries and Irish Officials.
4:45 PM – 5:00 PM	Closing Session: Key Takeaways and Next Steps	**Facilitator:** OECD Representative. **Closing Remarks:** Final reflections and thanks from the Irish Department of Education and OECD.

Annex B. Digital education initiatives mentioned in this paper

This annex includes tables presenting all digital education initiatives referred to in this paper. The tables do not represent an exhaustive list of countries' digital education initiatives. Initiatives appear with the same name as referred to in the paper. They are organised by countries and accompanied by brief descriptions.

Table B.1. Digital education initiatives in Estonia

Name of initiative	Brief description
AI Leap 2025	The **AI Leap 2025 initiative**, launched by the Estonian government, aims to integrate artificial intelligence tools into schools nationwide by 2025, enhancing digital education through strategic partnerships with leading AI developers and targeted teacher training.
DigiEfekt Study	The **DigiEfekt** project, conducted by the University of Tartu from 2020 to 2023, examined the impact of digital learning materials on Estonian basic school students' cognitive and non-cognitive skills, providing data-driven recommendations for educators, developers, and policymakers.
Digital Competency Framework	The **Digital Competence Framework** outlines Estonia's approach to integrating digital skills into education, providing structured guidelines for both students and teachers to develop, assess, and apply digital competencies across all levels of learning.
Education 2035	The **Education Strategy 2021–2035** outlines Estonia's long-term vision for education, focusing on equal opportunities, lifelong learning, and the development of digital and green competencies to prepare learners for future societal and labour market needs.
eKool	**eKool** is a digital school management platform in Estonia that connects students, parents, teachers, and school administrators, streamlining access to academic information, communication, and administrative processes in general education.
Lifelong Learning Strategy 2020	The **Estonian Lifelong Learning Strategy 2020** set out the country's educational priorities for 2014–2020, focusing on modernising learning approaches, strengthening teacher competencies, aligning education with labour market needs, promoting digital education and sustainability, and ensuring equitable access to lifelong learning.
Estonia's national curriculum	Estonia's **National Curricula** for basic and upper secondary schools, developed by the Ministry of Education and Research, establish the educational standards and learning objectives across various subjects, ensuring a cohesive and comprehensive framework for general education nationwide.
ProgeTiger programme	**ProgeTiger** is a national initiative in Estonia that supports the development of digital, programming, and robotics skills from preschool to secondary education, by providing resources, training, and curriculum integration to foster technological creativity and competence among students.
Teacher Education Academy	The **Teachers Education Academy** provides international professional development for educators through hands-on training, study visits, and seminars, with a focus on digital innovation, inclusive education, and cross-cultural collaboration.
Tiger Leap programme	The **Tiger Leap program**, launched in 1996, modernised Estonia's education system by introducing ICT infrastructure and teacher training, laying the groundwork for the country's digital transformation.

Table B.2. Digital education initiatives in France

Name of initiative	Brief description
Cadre de Référence des Compétences Numériques pour l'Éducation (CRCN-Édu)	The **Cadre de Référence des Compétences Numériques pour l'Éducation (CRCN-Édu)** is a framework that defines 16 professional digital competencies across five domains—professional environment, digital resources, teaching and learning, learner diversity and autonomy, and student digital skills—designed to guide educators in effectively integrating digital tools into their teaching practices.

Digital Schools Plan (Plan numérique pour l'éducation)	The **2015 Digital Plan for Education** aimed for the nationwide deployment of educational resources and mobile digital equipment in public primary schools and both public and private under contract middle schools based on calls for projects.
Digital Strategy for Education 2023–2027	The **2023–2027 Digital Strategy for Education** aims to strengthen digital education by fostering collaboration between national and local stakeholders, enhancing students' digital skills, supporting teachers with training and resources, and improving digital infrastructure to ensure secure, accessible, and environmentally responsible technology use in schools.
Direction Du Numérique Pour L'éducation	The **Direction du Numérique pour l'Éducation (DNE)** is a department within France's Ministry of National Education, responsible for driving and supporting the digital transformation of the educational system by overseeing both the technical infrastructure and the development of digital educational policies and tools.
École Numérique Rurale (ENR)	The "**École numérique rurale**" (Digital Rural School) plan aims to reduce the digital divide by equipping rural primary schools with digital tools, enhancing educational practices, and fostering innovation in teaching methods.
EduPilote	EduPilote is a digital dashboard developed which will monitor and coordinate the implementation of digital education initiatives across public middle and high schools, focusing on usage, equipment, and teacher training.
Élaine	**Élaine** (Évaluation Longitudinale des Activités liées au Numérique Éducatif) is a scientific study assessing the impact of the 2015 Digital Plan for Education. In particular the study analyses the impact of the Plan's provision of digital equipment to students on educational outcomes, teaching practices, and school management.
Numérique Inclusif	The École Inclusive initiative ensures that all students, including those with special educational needs, receive appropriate support to learn and succeed together in mainstream classrooms.
Pix	**Pix** is a national digital platform that enables individuals—including students and the general public—to assess, develop, and certify their digital competencies through adaptive learning and standardised benchmarks.
Pix+ Édu	**Pix+ Édu** is a dedicated branch of the Pix platform that supports education professionals in evaluating and strengthening their digital teaching competencies, in alignment with France's professional digital skills framework.
Territoires Numériques Éducatifs	The **Territoires Numériques Éducatifs (TNE)** initiative is a national programme in France that aims to enhance educational equity and digital inclusion by providing schools in selected departments with digital equipment, innovative educational resources, teacher training, and support for families to ensure effective pedagogical continuity.
Territories of Educational Priority (REP)	L'**Éducation Prioritaire** is a national policy in France aimed at reducing educational inequalities by providing additional resources, support, and targeted interventions to schools in socially and economically disadvantaged areas.

Table B.3. Digital education initiatives in Ireland

Name of initiative	Brief description
Céim	The **Céim** Standards for Initial Teacher Education set out the professional requirements and programme standards for the design, delivery, and accreditation of initial teacher education in Ireland.
Digital Learning Framework (DLF)	Ireland's **Digital Strategy for Schools to 2027** provides a national framework to integrate digital technologies in education, focusing on infrastructure, teacher development, and equitable access.
Digital Strategy for Schools	Ireland's **Digital Strategy for Schools to 2027** provides a national framework to integrate digital technologies in education, focusing on infrastructure, teacher development, and equitable access.
ICT Grant Scheme	The **Grant Scheme for ICT Infrastructure** provides funding to schools to enhance their digital capacity by supporting the purchase of ICT equipment and related infrastructure for teaching and learning.
Implementation Plan for the Digital Strategy for Schools	The **Implementation Plan for the Digital Strategy for Schools** outlines the actions, timelines, and responsibilities required to achieve the strategic objectives for integrating digital technologies in education.
Literacy, Numeracy and Digital Literacy Strategy 2024-2033	The **Literacy, Numeracy and Digital Literacy Strategy 2024–2033** sets out a national framework to strengthen core skills from early childhood to young adulthood through inclusive, lifelong learning supports.
Literacy, Numeracy and Digital Literacy Strategy 2024–2033: Implementation Plan to 2028	Ireland's **Literacy, Numeracy and Digital Literacy Strategy 2024–2033: Implementation Plan to 2028** outlines targeted actions to improve core and digital literacy skills for all learners from birth to young adulthood through coordinated support for educators, families, and school system.
Junior Cycle English curriculum	The **Junior Cycle English curriculum** in Ireland focuses on developing students' literacy, critical thinking, and communication skills through a learner-centred, outcomes-based approach that integrates reading, writing, and oral language across a three-year programme.
Key Competencies in Senior Cycle framework	The **Key Competencies in Senior Cycle framework** defines seven core competencies integrated across subjects to support students' holistic development and preparedness for life beyond school.
Leaving Certificate Computer Science	**Leaving Certificate Computer Science** fosters creativity, problem-solving, and collaboration, while equipping students with core computing skills and an understanding of technology's role in society and the environment.

Delivering Equality of Opportunity in Schools (DEIS)	**DEIS** – Delivering Equality of Opportunity in Schools – is a policy framework that provides targeted supports to address educational disadvantage and promote equity in learning outcomes for students in under-resourced communities.
Oide's Technology in Education (TiE)	**Oide Technology in Education** supports the effective use of digital technologies in teaching, learning, and school leadership by providing professional development, resources, and infrastructure guidance to educators and schools.
Primary Curriculum Framework	The **Primary Curriculum Framework** (2023) outlines a modernised vision for Irish primary and special education, promoting inclusive, child-centred learning through broad curriculum areas and key competencies to ensure continuity and progression.
Schedule of Senior Cycle Subjects for Redevelopment	The **Schedule of Senior Cycle Subjects for Redevelopment** outlines Ireland's phased approach to modernising senior cycle curricula to ensure relevance, coherence, and alignment with students' evolving learning needs.
Schools Broadband Programme	The **Schools Broadband Programme** delivers centrally managed, secure, and filtered broadband connectivity to schools, supporting the integration of digital technologies in teaching, learning, and school administration.
Scoilnet,	**Scoilnet** is the official education portal providing curated digital resources, curriculum content, and collaborative tools to support teaching and learning across all levels of the education system.
STEM Education Policy Statement 2017–2026	The **STEM Education Policy Statement 2017–2026** outlines a national vision to enhance engagement and achievement in science, technology, engineering, and mathematics through high-quality teaching, learning, and system-wide collaboration.
Webwise programme	**Webwise** is a national Internet safety initiative that offers comprehensive resources and educational support to help students, educators, and parents promote safe, ethical, and informed use of digital technologies.

Table B.4. Digital education initiatives in Sweden

Name of initiative	Brief description
2018 revision of school curricula	The 2018 revision of Sweden's **Curriculum for the Compulsory School, Preschool Class and School-Age Educare** (Lgr22) formally reinforced media and information literacy by mandating students' ability to critically assess sources, engage responsibly with digital content, and understand the role of media in a democratic society.
National Digitalisation Strategy for the School System	Sweden's **National Digitalisation Strategy for the School System** (2017) aims to enhance digital competence among students and educators, ensure equitable access to digital tools and infrastructure, and promote research on digitalisation's impact on education.
SALAR (Swedish Association of Local Authorities and Regions)	The Swedish **Association of Local Authorities and Regions (SALAR)** represents and supports municipalities and regions in Sweden, providing expertise, advocacy, and coordination across sectors including education, health, and digitalisation.
Skoldigistöd (OECD, 2023[37])	**Skoldigistöd** is a national initiative led by the Swedish Association of Local Authorities and Regions (SKR) that provides strategic support, resources, and best practices to municipalities and schools for the effective and equitable integration of digital tools in education.

References

Andric, A. (2023), *Sweden – National Digitalisation Strategy for the School System 2023-2027 | Digital Skills and Jobs Platform*, https://digital-skills-jobs.europa.eu/en/actions/national-initiatives/national-strategies/sweden-national-digitalisation-strategy-school-0 (accessed on 10 April 2025). [22]

Cukurova, M. and R. Luckin (2019), *Designing Educational Technologies in the Age of AI: A Learning Sciences-Driven Approach*, https://discovery.ucl.ac.uk/id/eprint/10078866/. [10]

Daniel L. Stufflebeam (2001), "Evaluation Models", *New Directions for Evaluation* 89, https://doi.org/10.1002/ev.3 (accessed on 13 April 2025). [8]

Department of Education and Youth (2024), *Webwise*, https://www.webwise.ie/ (accessed on 12 December 2024). [31]

Department of Education and Youth (2022), *Digital Strategy for Schools to 2027*, Government of Ireland, https://www.gov.ie/en/publication/69fb88-digital-strategy-for-schools/. [5]

Department of Education Youth and Department of Children (2023), *Ireland's Literacy, Numeracy and Digital Literacy Strategy 2024–2033: Every Learner from Birth to Young Adulthood*. [35]

Digital Skills & Jobs Platform (2022), *Estonia - Education Strategy 2021–2035.*, https://digital-skills-jobs.europa.eu/en/actions/national-initiatives/national-strategies/estonia-education-strategy-2021-2035 (accessed on 12 December 2024). [13]

Direction de L'Évaluation de la Prospective et de la performance (DEPP) (2017), *Multi-dimensional evaluation of the impact of mobile digital equipment on student learnings: Preliminary results of the effects of the 2015 Digital Plan*. [20]

Education Estonia (2024), *Enhancing learning through digital integration*, https://www.educationestonia.org/digital-learning-study/. [16]

Education Estonia (2020), *How did Estonia become a new role model in digital education?*, https://www.educationestonia.org/how-did-estonia-become-a-new-role-model-in-digital-education/. [15]

Éduscol (2024), *Éducation prioritaire - repères historiques | éduscol | Ministère de l'Éducation Nationale, de l'Enseignement supérieur et de la Recherche | Dgesco*, https://eduscol.education.fr/3568/education-prioritaire-reperes-historiques (accessed on 14 April 2025). [34]

e-Estonia (2025), *Estonia's groundbreaking national initiative: AI Leap programme to bring AI tools to all schools*, https://e-estonia.com/estonia-announces-a-groundbreaking-national-initiative-ai-leap-programme-to-bring-ai-tools-to-all-schools/ (accessed on 10 April 2025). [14]

European Commission (2024), *Media Literacy and Safe Use of New Media - Sweden*, https://national-policies.eacea.ec.europa.eu/youthwiki/chapters/sweden/68-media-literacy-and-safe-use-of-new-media. [23]

Fraillon, J. et al. (2020), *Preparing for Life in a Digital World: The IEA International Computer and Information Literacy Study (ICILS) 2018 International Report*, Springer Cham, https://link.springer.com/book/10.1007/978-3-030-38781-5. [25]

Golden, G. (2020), "Education policy evaluation: Surveying the OECD landscape", *OECD Education Working Papers*, No. 236, OECD Publishing, Paris, https://doi.org/10.1787/9f127490-en. [6]

Hallie Preskill et al. (2013), *Evaluating Complexity: Propositions for Improving Practice*, https://www.fsg.org/resource/evaluating-complexity/ (accessed on 13 April 2025). [7]

IEA (2024), *International Computer and Information Literacy Study 2023*, https://www.iea.nl/studies/iea/icils/2023. [38]

Joint Research Centre (2023), *Unpacking the impact of digital technologies in Education*, European Commission, https://publications.jrc.ec.europa.eu/repository/bitstream/JRC132998/JRC132998_01.pdf. [9]

Koehler, M. (2013), "What is Technological Pedagogical Content Knowledge (TPACK)?", *Journal of Education*, Vol. 193/3, pp. 13-19, https://doi.org/10.1177/002205741319300303. [33]

Ministère de l'Éducation nationale et de la Jeunesse (2024), *Direction du numérique pour l'éducation (DNE)*, https://www.education.gouv.fr/direction-du-numerique-pour-l-education-dne-9983 (accessed on 11 December 2024). [19]

Ministère de l'Éducation nationale et de la Jeunesse (2023), *Numérique pour l'éducation 2023-2027 - La vision stratégique d'une politique publique partagée*, https://www.education.gouv.fr/media/120418/download (accessed on 22 April 2025). [21]

Ministère de l'Éducation nationale, D. (n.d.), *Territoires Numériques Éducatifs - Territoires Numériques Éducatifs*, https://tne.reseau-canope.fr/ (accessed on 13 April 2025). [18]

Ministry of Education and Research of Estonia (2021), *Education strategy 2021-2035*. [17]

National Council for Curriculum and Assessment (NCCA) (2023), *Primary Curriculum Framework for Primary and Special Schools*. [30]

OECD (2025), *Resourcing School Education: Policies for the Digital Transformation of Education and Future-Readiness of Teachers [website]*, https://www.oecd.org/en/about/projects/resourcing-school-education-for-the-digital-age.html (accessed on 13 April 2025). [32]

OECD (2024), *"Students, digital devices and success"*, OECD Publishing, https://doi.org/10.1787/9e4c0624-en. [2]

OECD (2024), *Managing screen time: How to protect and equip students against distraction*, OECD Publishing, Paris, https://doi.org/10.1787/7c225af4-en. [1]

OECD (2023), *"Sweden", in Country Digital Education Ecosystems and Governance: A Companion to Digital Education Outlook 2023*, OECD Publishing, https://doi.org/10.1787/906134d4-en. [26]

OECD (2023), *Country Digital Education Ecosystems and Governance: A Companion to Digital Education Outlook 2023*, OECD Publishing, Paris, https://doi.org/10.1787/906134d4-en. [37]

OECD (2023), *Digital Education Outlook 2023: Pushing the Frontiers with Artificial Intelligence, Blockchain and Robots*, OECD Publishing, https://www.oecd-ilibrary.org/education/digital-education-outlook-2023_9789264311992-en. [27]

OECD (2023), *PISA 2022 Results (Volume I and II) - Country Notes: Ireland*, OECD Publishing, https://www.oecd.org/en/publications/pisa-2022-results-volume-i-and-ii-country-notes_ed6fbcc5-en/ireland_01173012-en.html. [4]

OECD (2023), *PISA 2022 Results (Volume II): Learning During – and From – Disruption*, PISA, OECD Publishing, Paris, https://doi.org/10.1787/a97db61c-en. [3]

OECD (2023), *Shaping Digital Education: Enabling Factors for Quality, Equity and Efficiency*, OECD Publishing, https://doi.org/10.1787/bac4dc9f-en. [11]

Oide Technology in Education (2024), *Digital Learning Framework - Planning Resources*, https://www.dlplanning.ie/ (accessed on 12 December 2024). [28]

Oide Technology in Education (2024), *Digital Learning Framework - Planning Resources*, https://www.dlplanning.ie/ (accessed on 14 April 2025). [29]

Skolverket (2022), *Follow-Up Report on Digitalisation in Schools*, https://www.skolverket.se/getFile?file=9385. [24]

The Teaching Council (2020), *Céim: Standards for Initial Teacher Education*, https://www.teachingcouncil.ie/assets/uploads/2023/08/ceim-standards-for-initial-teacher-education.pdf. [39]

Tiger Leap Foundation (2007), *Tiger Leap*. [12]

University of Tartu (2024), *Digiefekt Study*, https://www.educationestonia.org/digital-learning-study/. [36]

Notes

[1] While PISA provides valuable insights into the relationship between technology use and student performance, it is also important to recognise some limitations. Although PISA data empirically identifies consistent international patterns of correlations between student performance and the use of technology, PISA cannot establish causality due to its cross-sectional design. Furthermore, PISA data on the use of digital technology is based on self-reports by students, teachers, and school leaders, rather than direct observations of classroom practices and its student questionnaires focus on a limited set of activities and digital devices and the frequency with which they are used. To allow for a more comprehensive assessment of the effect of technology integration on student learning, other methodological approaches are required, such as randomised control trials. Recent research, such as the Estonian DigiEfekt Study, is a good example of the kind of approaches countries might pursue to examine, in more detail, how technology is used in educational settings (more information related to this study is available on this document).

[2] The International Computer and Information Literacy Study (ICILS) is a computer-based international assessment developed by the International Association for the Evaluation of Educational Achievement (IEA). It evaluates the ability of Grade 8 students (approximately 13.5 years old) to use information and communication technology (ICT) effectively for study, work, and life in a digital world. The test focuses on two main domains: 1) Computer and Information Literacy (CIL): this measures skills in using computers to investigate, create, and communicate, including tasks such as gathering, managing, and producing information digitally; and 2) Computational Thinking (CT): this assesses the ability to conceptualise problems and develop solutions using computational approaches. The ICILS also collects contextual data through questionnaires for students, teachers, school principals, and ICT coordinators to understand the environments influencing ICT education. First conducted in 2013, the study has been repeated in 2018, 2023 and is planned for 2028. ICILS provides policymakers with insights into digital literacy education across countries and tracks progress over time (IEA, 2024[38]).

[3] For a list of **Estonia**'s digital education initiatives please see Table B.1 in Annex B.

[4] In its first phase (from September 2025), the programme will involve 20,000 high school students (grades 10–11) and 3,000 teachers, with plans to expand to vocational schools and new incoming 10th-grade students in 2026 to add 38,000 students and 2,000 teachers to the pool. The initiative lies on a public-private cooperation where Estonia's Ministry of Education and Research is working with global AI developers to reach the programme's goal of providing free access to AI applications and training (e-Estonia, 2025[14]).

[5] For a list of **France**'s digital education initiatives please see Table B.2 in Annex B.

[6] The Direction Du Numérique Pour L'éducation the directorate responsible for integrating digital technology into the educatoinal system within the “Ministère de l’Education nationale, de l’Enseignement supérieur et de la Recherche”. The directorate promotes the use of digital tools in teaching and learning, supports educational innovation, and ensures access to digital resources.

[7] For a list of **Sweden**’s digital education initiatives please see Table B.4 in Annex B.

[8] For a list of **Estonia**’s digital education initiatives please see Table B.3 in Annex B.

[9] Céim, the Teaching Council’s standards for initial teacher education programmes, outlines the key components of ITE programmes, focusing on integrating foundation studies, professional studies, and school placements with digital skills essential for teaching and assessment. The programme aims to develop reflective practitioners, equipping them to use digital technologies effectively. Graduate teacher standards emphasise professional values, ongoing learning, understanding digital literacy and ethical technology use, and using digital tools for teaching and assessment. The core ITE elements promote inclusive education, integrating digital technologies to support diverse learning needs. Digital literacy is embedded throughout the programme, ensuring teachers are prepared to leverage technology in their practice (The Teaching Council, 2020[39]).

Drawing on insights from an International Peer-Learning Event hosted by the OECD and the Irish Department of Education, this document summarises key evidence and insights presented by representatives from Estonia, France, and Sweden as well as practical considerations for Irish policymakers. Key issues addressed include the balance between educational benefits and disadvantages, the importance of equitable access to technology, and the need for teacher competence and professional development. This document briefly highlights the practical options discussed during the event for Irish policymakers to foster a continuous reflection about how to enhance its digital education initiatives, ensuring they are effective, inclusive, and aligned with broader educational goals.

Connect with us:

edu.contact@oecd.org

https://oecdedutoday.com/

https://www.oecd-ilibrary.org/education

x @OECDEduSkills

in OECD Education and skills

ig @oecd_education_skills

OECD
BETTER POLICIES FOR BETTER LIVES